

MATH 231 – Midterm 1 Cheat Sheet

Homework 1 and Homework 2 material ■ vectors, complex numbers, derivatives ■ Kevin Bedoya, Fall 2026

1 Notation

$\mathbf{u}, \mathbf{v}, \mathbf{x}$	vectors (bold)
α, β, k, t	scalars : real numbers
$[u_1, u_2]$	vector of $\mathbb{R}^2$ with components u_1, u_2
$\mathbb{R}^n$	all lists $[v_1, \dots, v_n]$ of n reals
$\mathbf{0}$	zero vector $[0, \dots, 0]$
$\mathbf{e}_i$	standard basis vector: 1 in slot i , 0 else
$\mathbf{1}$	all-ones vector $[1, 1, \dots, 1]$
$\langle \mathbf{u}, \mathbf{v} \rangle, \mathbf{u} \cdot \mathbf{v}$	dot product (a <i>scalar</i>)
$\ \mathbf{v}\ , \ \mathbf{v}\ _2$	Euclidean norm (length)
$\ \mathbf{v}\ _1, \ \mathbf{v}\ _\infty$	Manhattan; Chebyshev norm
$\ \mathbf{v}\ _p$	the p -norm, $p \geq 1$
$d_p(\mathbf{x}, \mathbf{y})$	distance $\ \mathbf{x} - \mathbf{y}\ _p$
$\hat{\mathbf{u}}$	unit vector $\mathbf{u}/\ \mathbf{u}\ $
$\cos(\mathbf{u}, \mathbf{v})$	cosine similarity
$\theta \in [0, \pi]$	angle between two vectors
$\mathbf{u} \perp \mathbf{v}$	orthogonal: $\langle \mathbf{u}, \mathbf{v} \rangle = 0$
$\text{comp}_{\mathbf{v}}(\mathbf{u})$	component of $\mathbf{u}$ along $\mathbf{v}$ (<i>scalar</i>)
$\text{proj}_{\mathbf{v}}(\mathbf{u})$	projection of $\mathbf{u}$ onto $\mathbf{v}$ (<i>vector</i>)
$\text{span}\{\mathbf{u}, \mathbf{v}\}$	all combinations $\alpha\mathbf{u} + \beta\mathbf{v}$
$\dim V$	dimension of V
$\in, \notin$	is / is not an element of
$\subseteq, \cap, \cup$	subset; intersection; union
$\{\mathbf{x} : P\}$	the set of all $\mathbf{x}$ with property P
$\forall, \exists$	for all; there exists
$\Rightarrow, \Leftrightarrow$	implies; if and only if
$\sum_{i=1}^n a_i$	$a_1 + a_2 + \dots + a_n$
$\max_i v_i $	the largest $ v_i $
$f : A \rightarrow B$	signature: inputs from A , outputs in B
$f'(x), \frac{d}{dx}$	derivative
$\partial f / \partial x$	partial derivative
∇f	gradient $[\partial f / \partial x, \partial f / \partial y]$
$i; \mathbb{C}$	$i^2 = -1$; the complex numbers
$\text{Re } z, \text{Im } z$	real part, imaginary part
$\bar{z}; z $	conjugate; modulus

Signatures. $\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R}$ is *unary*; $+$: $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $\langle \cdot, \cdot \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ are *binary*; scalar multiplication is $\mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Vector vs. scalar quantity. A vector quantity has a magnitude *and* a direction (velocity, force, displacement, acceleration, momentum, a gradient). A scalar quantity has magnitude only (speed, mass, work, temperature, distance travelled).

2 Vector formulas

2.1 Arithmetic [V1 §7–§9]

Componentwise, in any $\mathbb{R}^n$:

$$\mathbf{u} \pm \mathbf{v} = [u_1 \pm v_1, \dots, u_n \pm v_n],$$

$$\alpha \mathbf{v} = [\alpha v_1, \dots, \alpha v_n].$$

$$-\mathbf{v} = (-1)\mathbf{v} \text{ and } \mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}).$$

- Arrow** from tail P to head Q : the vector $Q - P$ (*head minus tail*).
- Equality**: $\mathbf{u} = \mathbf{v}$ iff every component agrees. Equal norms are *not* enough.
- $\alpha > 0$ keeps the direction, $\alpha < 0$ reverses it, $|\alpha|$ scales the length; $0\mathbf{v} = \mathbf{0}$.

The eight laws [V1 §8.4], for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars α, β :

- A1 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
A2 $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
A3 $\mathbf{u} + \mathbf{0} = \mathbf{u}$
A4 $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
S1 $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$
S2 $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$
S3 $\alpha(\beta\mathbf{u}) = (\alpha\beta)\mathbf{u}$
S4 $1\mathbf{u} = \mathbf{u}$

! Subtraction is neither commutative nor associative: $\mathbf{u} - \mathbf{v} = -(\mathbf{v} - \mathbf{u})$.

2.2 Norms and distance [V1 §2, V3 §14]

$$\|\mathbf{v}\|_2 = \sqrt{\sum v_i^2}, \quad \|\mathbf{v}\|_2^2 = \langle \mathbf{v}, \mathbf{v} \rangle,$$

$$\|\mathbf{v}\|_1 = \sum |v_i|, \quad \|\mathbf{v}\|_\infty = \max_i |v_i|,$$

$$\|\mathbf{v}\|_p = \left(\sum |v_i|^p \right)^{1/p} \quad (p \geq 1).$$

- Scaling**: $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$, because $\sqrt{\alpha^2} = |\alpha|$.
- Unit vector**: $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$ for $\mathbf{u} \neq \mathbf{0}$; then $\mathbf{u} = \|\mathbf{u}\|\hat{\mathbf{u}}$. Length L along $\mathbf{u}$: $L\hat{\mathbf{u}}$; opposite: $-L\hat{\mathbf{u}}$.
- Triangle inequality**: $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, with equality iff one is a multiple $\alpha \geq 0$ of the other.
- Reverse**: $\|\|\mathbf{u}\| - \|\mathbf{v}\|\| \leq \|\mathbf{u} - \mathbf{v}\|$.
- Ordering** in $\mathbb{R}^n$:

$$\|\mathbf{v}\|_\infty \leq \|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq n\|\mathbf{v}\|_\infty;$$

$$\text{also } \|\mathbf{v}\|_2 \leq \sqrt{n}\|\mathbf{v}\|_\infty \text{ and } \|\mathbf{v}\|_1 \leq \sqrt{n}\|\mathbf{v}\|_2.$$

- Unit circles** $\{\|\mathbf{v}\|_p = 1\}$: $p = 1$ a diamond (corners $(\pm 1, 0), (0, \pm 1)$); $p = 2$ the round circle; $p = \infty$ a square (corners $(\pm 1, \pm 1)$).
- Which metric?** d_1 : taxi on a street grid. d_2 : drone in a straight line. d_∞ : crane, both motors at once.
- Compare square roots by squaring: $\sqrt{40} < 7 = \sqrt{49}$.

2.3 Dot product and angles [V2 §10–§11, V3 §13]

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i = \|\mathbf{u}\|\|\mathbf{v}\| \cos \theta,$$

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\|\mathbf{v}\|} \in [-1, 1],$$

$$\theta = \arccos(\cos(\mathbf{u}, \mathbf{v})) \in [0, \pi].$$

$\cos(\mathbf{u}, \mathbf{v})$	angle	the pair is
1	0	parallel: $\mathbf{v} = \alpha\mathbf{u}, \alpha > 0$
(0, 1)	acute	none of the three
0	$\pi/2$	orthogonal
(-1, 0)	obtuse	none of the three
-1	π	antiparallel: $\mathbf{v} = \alpha\mathbf{u}, \alpha < 0$

Cosine similarity ignores length: $\cos(\alpha\mathbf{u}, \mathbf{v}) = \cos(\mathbf{u}, \mathbf{v})$ for $\alpha > 0$, but $\cos(-\mathbf{u}, \mathbf{v}) = -\cos(\mathbf{u}, \mathbf{v})$.

The dot-product laws [V2 §11.8]:

- D1 $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
D2 $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
D3 $\langle \mathbf{u}, \alpha\mathbf{v} \rangle = \alpha\langle \mathbf{u}, \mathbf{v} \rangle$
D4 $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2 \geq 0, = 0$ only for $\mathbf{u} = \mathbf{0}$

With D1, D2 and D3 also work in the first slot; and $\langle \mathbf{x}, \mathbf{0} \rangle = 0$ for every $\mathbf{x}$.

Expansions, from D1–D4. The third line is the

parallelogram law, the fourth *polarization*:

$$\|\mathbf{u} \pm \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \pm 2\langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2,$$

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2,$$

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2,$$

$$\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2).$$

Cauchy–Schwarz: $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|$, with equality iff one is a multiple of the other.

! No cancellation: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle$ gives only $\mathbf{u} \perp (\mathbf{v} - \mathbf{w})$, not $\mathbf{v} = \mathbf{w}$.

2.4 Component and projection [V2 §12]

For $\mathbf{v} \neq \mathbf{0}$; the subscript is the vector projected onto.

$$\text{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\|\mathbf{v}\|} = \|\mathbf{u}\| \cos \theta \quad (\text{a scalar}),$$

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\|\mathbf{v}\|^2} \mathbf{v} = \text{comp}_{\mathbf{v}}(\mathbf{u}) \hat{\mathbf{v}} \quad (\text{a vector}).$$

- Orthogonal piece** $\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})$ is $\perp \mathbf{v}$, and $\mathbf{u} = \text{proj}_{\mathbf{v}}(\mathbf{u}) + (\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u}))$.
- $\|\mathbf{u}\|^2 = \|\text{proj}_{\mathbf{v}}(\mathbf{u})\|^2 + \|\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})\|^2$, so $\|\text{proj}_{\mathbf{v}}(\mathbf{u})\| \leq \|\mathbf{u}\|$.
- $\text{proj}_{\alpha\mathbf{v}}(\mathbf{u}) = \text{proj}_{\mathbf{v}}(\mathbf{u})$ for $\alpha \neq 0$; the component flips sign when $\alpha < 0$.
- $\text{proj}_{\mathbf{v}}(\text{proj}_{\mathbf{v}}(\mathbf{u})) = \text{proj}_{\mathbf{v}}(\mathbf{u})$, and projection is linear in what is projected.
- In general $\text{proj}_{\mathbf{v}}(\mathbf{u}) \neq \text{proj}_{\mathbf{u}}(\mathbf{v})$.

Work done by a constant force $\mathbf{F}$ over displacement $\mathbf{s}$: $W = \langle \mathbf{F}, \mathbf{s} \rangle = \text{comp}_{\mathbf{s}}(\mathbf{F}) \|\mathbf{s}\|$ (joules). For a fixed $\|\mathbf{F}\|$, W is largest when $\mathbf{F}$ points along $\mathbf{s}$, and $W = 0$ when $\mathbf{F} \perp \mathbf{s}$. On a ramp in direction $\mathbf{d}$, $\text{proj}_{\mathbf{d}}(\mathbf{G})$ pulls along the ramp and $\mathbf{G} - \text{proj}_{\mathbf{d}}(\mathbf{G})$ presses into it.

2.5 Particular to $\mathbb{R}^n$ [V5 §24]

- $\langle \mathbf{e}_i, \mathbf{v} \rangle = v_i$; $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = 1$ if $i = j$, else 0; $\|\mathbf{e}_i - \mathbf{e}_j\|_2 = \sqrt{2}$ for $i \neq j$.
- $\|\mathbf{1}\|_1 = n$, $\|\mathbf{1}\|_2 = \sqrt{n}$, $\|\mathbf{1}\|_\infty = 1$, $\cos(\mathbf{1}, \mathbf{e}_i) = 1/\sqrt{n}$.
- $\text{proj}_{\mathbf{1}}(\mathbf{x}) = \bar{x}\mathbf{1}$, where $\bar{x}$ is the mean of the entries.
- Documents** as word-count vectors: cosine similarity compares topic and ignores length; Euclidean distance is thrown off by length (a document pasted in twice has $\cos = 1$ but is far away).

2.6 Exact values for angles

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	π
$\cos$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	-1
$\sin$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	0

Degrees: $\pi/6 = 30^\circ$, $\pi/4 = 45^\circ$, $\pi/3 = 60^\circ$, $\pi/2 = 90^\circ$, $2\pi/3 = 120^\circ$, $3\pi/4 = 135^\circ$, $\pi = 180^\circ$. For $-\theta$, cosine is unchanged and sine flips sign.

2.7 Quick checks

- A cosine outside $[-1, 1]$ means an arithmetic error.
- A norm is never negative, and is 0 only for $\mathbf{0}$.
- An orthogonal piece must satisfy $\langle \mathbf{v}, \mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u}) \rangle = 0$.
- Coordinates are confirmed only by substituting them back.
- $\text{comp}_{\mathbf{v}}(\mathbf{u})$ is a number; $\text{proj}_{\mathbf{v}}(\mathbf{u})$ is a vector. Answer with the right kind of object.

3 Definitions

Combinations and span [V4 §16]

A **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_k$ is $\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k$ with scalars α_i . Their **span** is the set of all of them:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_i \alpha_i \mathbf{v}_i : \alpha_i \in \mathbb{R} \right\}.$$

A span always contains $\mathbf{0}$ and is always a subspace. If $\mathbf{w} \in \text{span}\{\mathbf{u}, \mathbf{v}\}$, then $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{span}\{\mathbf{u}, \mathbf{v}\}$.

Linear independence [V4 §18]

$\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** when

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0} \Rightarrow \alpha_1 = \dots = \alpha_k = 0.$$

They are **linearly dependent** when some combination with the α_i not all zero gives $\mathbf{0}$; equivalently, one of them is a combination of the others.

- A pair in $\mathbb{R}^2$: dependent $\Leftrightarrow$ one is a multiple of the other $\Leftrightarrow D = u_1 v_2 - u_2 v_1 = 0$.
- A set containing $\mathbf{0}$ is dependent.
- Three or more vectors in $\mathbb{R}^2$ are always dependent.
- If $\mathbf{u}, \mathbf{v}$ are independent, so are $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$.

Basis, coordinates, dimension [V4 §16–§18]

A **basis** of a space V is a set of vectors that *spans* V and is *linearly independent*. Then every $\mathbf{w} \in V$ is a combination of the basis vectors in exactly one way; the coefficients are the **coordinates** of $\mathbf{w}$. Spanning gives existence of coordinates; independence gives uniqueness.

In $\mathbb{R}^2$ with $D \neq 0$, the coordinates in $\mathbf{w} = \alpha \mathbf{u} + \beta \mathbf{v}$ are

$$\alpha = \frac{w_1 v_2 - w_2 v_1}{D}, \quad \beta = \frac{u_1 w_2 - u_2 w_1}{D},$$

or solve the system by elimination. Always substitute back to check.

The **dimension** of V is the number of vectors in a basis: $\dim \mathbb{R}^n = n$; a line through $\mathbf{0}$ has dimension 1; $\{\mathbf{0}\}$ has dimension 0. The **standard basis** of $\mathbb{R}^n$ is $\mathbf{e}_1, \dots, \mathbf{e}_n$.

Subspaces [V4 §19]

A **subspace** of $\mathbb{R}^2$ is a subset S that is (1) non-empty, (2) closed under addition: $\mathbf{w}_1, \mathbf{w}_2 \in S \Rightarrow \mathbf{w}_1 + \mathbf{w}_2 \in S$, and (3) closed under scalar multiplication: $\mathbf{w} \in S \Rightarrow \gamma \mathbf{w} \in S$ for every scalar γ .

- Every subspace contains $\mathbf{0}$. If a set misses $\mathbf{0}$, it is not a subspace.
- The subspaces of $\mathbb{R}^2$ are exactly $\{\mathbf{0}\}$, the lines through the origin, and $\mathbb{R}^2$. In $\mathbb{R}^3$, add the planes through the origin.
- $S \cap T$ is a subspace. $S \cup T$ is one only if $S \subseteq T$ or $T \subseteq S$.
- **Not** subspaces: $x + y = 1$ (misses $\mathbf{0}$); $xy = 0$ and $y = x^2$ (not closed under +); $x, y \geq 0$ (not closed under $\gamma = -1$); integer points; the disc $x^2 + y^2 \leq 1$.
- To show a set is *not* a subspace, give explicit vectors (and a scalar) that break one condition.

Orthogonality [V2 §10, V5 §21–§23]

- $\mathbf{u}, \mathbf{v}$ are **orthogonal** ($\mathbf{u} \perp \mathbf{v}$) when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$. A set is **orthogonal** when each pair is, and **orthonormal** when it is orthogonal and every vector has norm 1.
- Nonzero, mutually orthogonal vectors are linearly independent. The converse fails: $[1, 0]$ and $[1, 1]$ are independent but not orthogonal.
- **Coordinates against an orthogonal basis**

need no system:

$$\mathbf{w} = \sum_j \frac{\langle \mathbf{b}_j, \mathbf{w} \rangle}{\|\mathbf{b}_j\|^2} \mathbf{b}_j = \sum_j \text{proj}_{\mathbf{b}_j}(\mathbf{w}).$$

- **Pythagorean identity**: if $\mathbf{u} \perp \mathbf{v}$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. For mutually orthogonal vectors, $\|\sum_j \alpha_j \mathbf{b}_j\|^2 = \sum_j \alpha_j^2 \|\mathbf{b}_j\|^2$; for orthonormal ones, $\sum_j \alpha_j^2$.
- **Making a basis orthogonal**: $\mathbf{v}' = \mathbf{v} - \text{proj}_{\mathbf{u}}(\mathbf{v})$ is $\perp \mathbf{u}$, and $\text{span}\{\mathbf{u}, \mathbf{v}'\} = \text{span}\{\mathbf{u}, \mathbf{v}\}$.
- $\mathbf{u}^\perp = \{\mathbf{x} : \langle \mathbf{u}, \mathbf{x} \rangle = 0\}$ is a subspace. If $\mathbf{w} \perp \mathbf{u}$ and $\mathbf{w} \perp \mathbf{v}$, then $\mathbf{w}$ is orthogonal to all of $\text{span}\{\mathbf{u}, \mathbf{v}\}$.

Norms and metrics [V3 §14.1, §15.1]

A **norm** satisfies, for all $\mathbf{u}, \mathbf{v}$ and scalars α : **N1** $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$; **N2** $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$; **N3** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$.

A **metric** satisfies, for all $\mathbf{x}, \mathbf{y}, \mathbf{z}$: **M1** $d(\mathbf{x}, \mathbf{y}) = 0$ iff $\mathbf{x} = \mathbf{y}$; **M2** $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$; **M3** $d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \geq d(\mathbf{x}, \mathbf{z})$.

Real-number facts for these proofs: $|ab| = |a||b|$ and $|a + b| \leq |a| + |b|$.

Proof habits. State what you assume. Name the law behind each step. “In components”: write $\mathbf{u} = [u_1, u_2]$ and use a fact about real numbers in each slot. “If and only if”: prove both directions, and say which one you are proving. End with a sentence saying what you have shown.

4 Derivatives

Definition and rules [VC1 §1–§2]

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Constants	$(c)' = 0$, $(cf)' = cf'$
Sum / difference	$(f \pm g)' = f' \pm g'$
Power	$(x^n)' = nx^{n-1}$ (n real)
Product	$(fg)' = f'g + fg'$
Quotient	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
Chain	$(f(g(x)))' = f'(g(x))g'(x)$

Derivatives to know [VC1 §2, §4]

f	f'	f	f'
e^x	e^x	$\sin x$	$\cos x$
$e^{g(x)}$	$g'(x)e^{g(x)}$	$\cos x$	$-\sin x$
$\ln x$	$1/x$	$\tan x$	$\sec^2 x$
$\ln(g(x))$	$g'(x)/g(x)$	$\sec x$	$\sec x \tan x$
a^x	$a^x \ln a$	$\csc x$	$-\csc x \cot x$
$\sqrt{x}$	$1/(2\sqrt{x})$	$\cot x$	$-\csc^2 x$

Exponential and logarithm laws [VC1 §3]

E1	$e^0 = 1$	L1	$\ln 1 = 0$
E2	$e^{a+b} = e^a e^b$	L2	$\ln(ab) = \ln a + \ln b$
E3	$e^{a-b} = e^a / e^b$	L3	$\ln(a/b) = \ln a - \ln b$
E4	$(e^a)^b = e^{ab}$	L4	$\ln(a^r) = r \ln a$
E5	$e^{-a} = 1/e^a$	L5	$\ln e = 1$

They are inverses: $e^{\ln x} = x$ for $x > 0$, and $\ln(e^x) = x$. So $a^x = e^{x \ln a}$, and $e^x > 0$ always.

- $e^{2x} + be^x + c = 0$: put $t = e^x$, solve the quadratic, keep only $t > 0$, then $x = \ln t$.
- In equations: combine with L2–L4, exponentiate, and reject any candidate that makes an argument of $\ln$ zero or negative.
- Implicit differentiation: $y = \ln x \Rightarrow e^y = x \Rightarrow e^y y' = 1 \Rightarrow y' = 1/x$.

Partial derivatives and the gradient [VC1 §5–§7]

$$\frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}.$$

To find $\partial f / \partial x$, differentiate in x and treat y as a constant; for $\partial f / \partial y$, the reverse. Its units are (units of f) / (units of x).

- **Gradient**: $\nabla f = [\partial f / \partial x, \partial f / \partial y]$, a vector. Evaluate it at a point to get a vector of numbers, then take its norm if asked.
- **Where $\nabla f = \mathbf{0}$** : solve both equations at once. Factor, and split into cases (for example $2y(x-1) = 0$ gives $y = 0$ or $x = 1$).
- **Estimating a change**: with y held fixed, $\Delta f \approx \frac{\partial f}{\partial x} \Delta x$; with x held fixed, $\Delta f \approx \frac{\partial f}{\partial y} \Delta y$.

5 Complex numbers

The basics [CN §2–§3]

$i^2 = -1$. A complex number is $z = a + bi$ with a, b real: $\text{Re } z = a$ and $\text{Im } z = b$. Both parts are *real*: $\text{Im}(4 - 3i) = -3$. Plot z at the point (a, b) of the complex plane. Every real number is complex ($b = 0$), and $z = w$ iff both parts agree.

Arithmetic [CN §4]

With $z = a + bi$ and $w = c + di$:

$$z \pm w = (a \pm c) + (b \pm d)i,$$

$$zw = (ac - bd) + (ad + bc)i,$$

$$\frac{z}{w} = \frac{z \bar{w}}{w \bar{w}} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2},$$

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Finish every quotient in the form $a + bi$.

Powers of i repeat with period 4:

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1.$$

So i^n depends only on the remainder of n divided by 4: $0 \mapsto 1$, $1 \mapsto i$, $2 \mapsto -1$, $3 \mapsto -i$. Also $(1+i)^2 = 2i$ and $(1-i)^2 = -2i$.

Conjugate and modulus [CN §4.3, §5]

$\bar{z} = a - bi$ is z reflected in the real axis, and $|z| = \sqrt{a^2 + b^2}$ is its distance from 0.

- $z\bar{z} = a^2 + b^2 = |z|^2$, a real number.
- $z + \bar{z} = 2\text{Re } z$ and $z - \bar{z} = 2i\text{Im } z$.
- $z \pm \bar{w} = \bar{z} \pm \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $\overline{\bar{z}} = z$.
- z is real iff $z = \bar{z}$; and $\text{Re } z = 0$ iff $z = -\bar{z}$.
- $|zw| = |z||w|$, $|\bar{z}| = |z|$, $|z + w| \leq |z| + |w|$.
- If $|z| = 1$, then $1/\bar{z} = z$.

Distance from z to w is $|z - w|$. With centre c and radius r : $|z - c| = r$ is a circle, $|z - c| \leq r$ a closed disc, and $|z - c| < r$ an open disc. Read the centre carefully: $|z + 2i| = |z - (-2i)|$ is centred at $-2i$.

The unit circle and Euler's formula [CN §6]

Points with $|z| = 1$ are $\cos \theta + i \sin \theta$, and Euler's formula gives

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

$$|e^{i\theta}| = 1, \quad |ze^{i\theta}| = |z|.$$

$e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{3\pi i/2} = -i$, $e^{2\pi i} = 1$, and $e^{i\pi} + 1 = 0$ is **Euler's identity**.

Multiplication as rotation [CN §7]

times	rotates the point	$(a, b) \mapsto$
i	90° counter-clockwise	$(-b, a)$
$-i$	90° clockwise	$(b, -a)$
-1	180°	$(-a, -b)$
$e^{i\theta}$	θ counter-clockwise	